

Active Inference BookStream #002.1 ~ Thomas Parr ~ Active Inference and

BookStream | Thomas Parr ~ Active Inference, Free Energy Principle... | 1:01:18

hello and welcome it's February 20th 7th

2023 we're here in active inference

bookstream number 2.1 with Thomas Parr

as well as Terry and Ali so thank you

all for joining this should be a very

fun discussion and let's go first by way

of introduction and beginning to Thomas

welcome and please

how did you come to write this textbook

in collaboration with pizulo and friston

well again thank you very much for

having me here

um

how did we come to write the textbooks

good question I think

um I think Giovanni was the the person

who really had the idea and pushed for

this that I think um Carl has already

previously been approached by

um by the editor who ended up managing

application of the book

um and

when Giovanni had the idea that this

would be a really useful contribution

um he and I had a number of discussions

about it both in in London and in Rome

where he's based

um and sort of put out a plan and and

sort of came up with something that we

hoped would be useful for the community

I suppose part of the part of the reason

why we thought it'd be a useful thing to

do is that active inference is a very

multi-disciplinary type field that draws

a lot from lots of different domains

and I think one of the one of the
struggles you know we all had when we
started out in it and I think certainly
what the students and uh people who are
interested in the topic say to us is
that it's often quite difficult to
trying to find all of these different
topics trying to find the right
background when they don't all come from
one field so putting it all together in
one book with a consistent notation
consistent language and trying to make
it complete we hoped would just make the
field a little bit more accessible
um to others who wanted to get involved
how did this dovetail or integrate with
your own Education and Research
directions like what phases or stages
were you moving through well this
discussion was happening
well I think so when we initially
started discussing it it must have been
probably the last year of my PhD and so
by that stage I'd spent uh three or four
sorry two to three years
um sort of getting to grips with active
inference
um having spent you know being
relatively recent in terms of learning
some of the key material myself I think
it was quite a good stage to be at in
terms of thinking about what I would
have wanted to know a few years
previously
um the actual writing of the book then
carried on over the next
year or two
um and so by that stage I was I was back

in clinical training as well
which I think is also a useful
perspective in terms of some of the
later chapters and thinking about
application of active infant site models
in the context of clinical conditions
and Psychiatry and neurology and those
sorts of areas
that's actually a really interesting
note how did the clinical experiences
give a perspective on the book or how
did the book or you're learning in these
abstract and theoretical domains give a
different Insight in the clinical
setting
it's a good question and I think you're
right to highlight the bi-directionality
of it that both the useful in in
different ways
um I think for me the clinical setting
is quite useful in terms of thinking you
know first of all in interacting the
different sort of different sort of um
audience so people who may be much more
skilled on the clinical side of things
maybe add less experience on the
mathematical computational side of
things but may still find the concepts
very useful so from that side of things
it's quite a useful exercise in being
able to communicate these things
um and then there's sort of the
interaction on the patient side of
things and trying to you know I think it
makes a big difference first of all in
interacting with patients has an impact
on how you might think about the
interesting questions and the things you

might want to use them to entrance for
but also having an understanding of
principles behind it helps you I think
formulate your questions about how
individual patients might present or the
reasons for that I think is a useful way
of of thinking about I mean I suppose on
two levels so one one in terms of
understanding

um

understanding particular sorts of
pathology in terms of the inferences
that might have gone wrong
um but also in terms of understanding
your own active influential processes
when you're trying to work out what's
going on yourself asking the right sort
of questions to gather the right sort of
information and and test those internal
hypotheses I think that's a very useful
way of thinking about
being a doctor

awesome yeah I like to say about that
the Bayesian optimal differential
diagnosis

Ali or Terry on any of these
thoughts do you have any questions or
comments you want to raise there or any
follow-ups

uh well yeah actually I just wanted to
ask uh what are the intended audience of
the book and what do you think are the
minimum prerequisites for going through
the whole book because I'm sure you'll
agree that even today the intersection
between the people who are well versed
in mathematics and the people who have a
solid background in say neurobiology are

quite narrow so

um

so what's the solution here and what um

did you think about

about the organizational structure of

the book that could accommodate various

uh audiences so

um yeah

again another really good question and I

think you're right to highlight that

that intersection is relatively narrow

and there are lots of lots of people who

I think

I struggle to feel involved in in those

kinds of discussions

um and you're absolutely right to say

that and I think you know what one of

the solutions is is the sorts of things

that you guys are doing in terms of

setting up these reading groups trying

to open that up trying to provide

educational resources I think that's

really very important

um in terms of the specific questions

about the book itself who's the audience

well

and you know what are the prerequisites

I think it sort of depends what you want

to get out of it and what your purpose

is in reading it

um so some aspects of the book I suppose

will appeal more to some audiences and

others to to other kinds of audience

um I would hope that the readership or

the audience is relatively broad because

we've tried to provide as much

background as we can on sort of the

mathematical and the neurobiological

side of things to try and

um

complement the skill set of whoever's

not in that intersection who comes to it

and who who may be need a bit more sort

of neurobiological grounding or who may

need a bit more mathematical and

Technical grounding

um but I think ultimately you like with

most textbooks you don't necessarily

need to come to it thinking I'm going to

understand everything in this book when

I come away from it some people may want

to

um but other people depending upon what

they want to get out of it they might

say actually what I really want to get

out of this is an understanding of

some of the the kind of message passing

as it's manifests in the brain whereas

other people might come to it and say

actually I already understand a bit

about that and what I want to get out of

this is to understand a bit more of the

technical detail

um you know if you're a robotic is

coming to this it probably doesn't

matter too much about how things are

implemented in the brain if you're a

neurobiologist coming to this you may

not need to know the details of the

implementation as long as you can

understand it conceptually and I hope

the organization particularly putting

things like dependencies into detailed

information at the back

tries to keep most of the text a bit

more accessible the early chapters being

relatively light on the technical
material and building up the concept
Concepts before then moving into the
more um technical aspects
about the uh chapters let's actually
just take a quick look at the chapters
um we're in the uh
Coda document that we use so perhaps you
could just say one or a few sentences on
each chapter what's the short of each
chapter what is it doing in this 10
chapter work
you're testing my memory now the titles
are there the cues are there titles
epistemic cues are there
okay
um well the overview is what it says on
the team it's a description of what's to
come in the subsequent chapters to try
and try and make orientate people
and the low range and high rate I think
again Giovanni has to take credit for
this idea of saying actually there's
quite a it's often difficult to know
where to begin when you're trying to
understand
um active interference in the free
energy principle and the Bayesian brain
and all those sorts of things because it
seems like there are so many potential
starting points
all of which ultimately I think get at
the same thing but knowing where you
begin is is often really quite difficult
um and so these two chapters were
designed to explicitly acknowledge that
um that difficulty
um so the idea of the low road is to say

well let's take a pragmatic perspective
um where we take ideas that have
developed in in Neuroscience based upon
our understanding of of the brain and of
psychology and models that have been
developed to explain those things
and let's take those to their conclusion
which is in a sense the free energy
principle or which is active inference
the idea of
um using our internal models both to
drive behavior and to understand what's
going on in the world around us
um the high road perspective was
designed to take a different approach to
that and instead of starting from what's
already been developed and the sort of
natural trajectory in the neurosciences
um the high road chapter was more saying
let's start from those principles as
much as we can
um and here was a tricky balance to try
and make sure that we we didn't go too
far in depth that we'd put people off by
the by the third chapter but enough that
people might get a bit of a conceptual
sense of where you might go with this
uh and and and here we we sort of
Started from the kind of physics of
sales organization the ideas that come
underneath that and how those ideas can
then feed back into this same idea of um
of accidents that the neurosciences in
Psychology seem to have arrived at
relatively naturally
suppose chapter four is when things
start getting a bit more technical
um and starts dealing with the specifics

that which I think most people would want to know if they were implementing these models themselves

um so trying to understand exactly how you construct an internal model how you specify how to write it down mathematically

um how you how you perform inference with those how you understand the message passing schemes

chapter 5 then then takes a step back and thinks well in the context of the brain in the context of neurobiology what do those things actually look like how do we understand those inferential processes with different kinds of generative models in terms of what we know about message passing and synaptic communication

and and that those those five chapters together really are the the first Partners book and the the subsequent five chapters really then take you through the idea of actually applying those ideas in particular domains

um so chapter six is sort of the overview chapter looking at the rest of it and it's really setting out if you wanted to sit down and design a model either conceptually your competition these are the steps you would have to go through to construct that

and then chapters uh seven and eight take you through that sort of recipe it for several examples building up different concepts and both the models designed in discrete time where you're formulating things as a sequence of

events

and models formulated in continuous time

we we work with continuous dynamical

systems and differential equations

chapter 9 as I mentioned before you know

one of my interests is in the clinical

domain and trying to understand what

sort of models people

um use to perform inference how those

can go wrong

um and part of part of the research on

that is thinking well how do we then fit

these models and use them to actually

draw inferences about what's going on

particularly individuals or groups of

individuals

um and then chapter 10 takes the

broadest perspective and and looks at

how that relates to a range another

series and and ideas both in the

neurosciences and Beyond

and then I'm sure dependencies speak for

themselves

awesome great overview let's come back

to our questions so

you laid out that the first half of the

book is about learning and the second

half of the book is about applying and

it begins with that recipe

so if chapter 6 is the recipe

as a chef and sous chef yourself what

else have you found was important in the

restaurant what is your full stock

active inference modeling look like and

maybe summarizing in six or going beyond

what was in six like what have you found

to be important as part of the modeling

Journey

especially with teams where people might have like different levels of domain specific knowledge or active knowledge

I like the restaurant analogy actually and I was thinking I suppose the first thing that anybody needs to do if they're setting up a restaurant or a model is thinking about who their customers are and what they might want um and so in the scientific domain you're thinking well what what sort of Journal am I planning on submitting to this to what sort of questions are the communities who read those um Journal articles interested in

um and I think that's a really important First Step so that you know what I'm actually trying to do with this model what's not such a good idea often although you can find interesting things by doing it is to say well I'm going to develop a model and work out what it's for later on you can end up sort of tying yourself up a little bit by doing that

although there is value in exploring us

I think we would all agree

and

then

I suppose in terms of in terms of ingredients It's relatively straightforward and deliberate deliberately so I suppose you know the idea of of these sorts of models is that they should be relatively accessible relatively easy to construct with um without having to make use of large

supercomputers or or anything like that
and a lot of the models that I develop
can be run on a laptop fairly
straightforwardly

um so you know obviously just picking
your your Computing language and how
you're going to construct that there are
now a couple of different options out
there

um Matlab being the one that I think is
most well developed but now lots of
people

um developing python-based
implementations of accidents and schemes
and particularly in the Markov decision
process Realm

um so that gives you I suppose that some
of the key ingredients but then it's
really sitting down and thinking well
how how would I now actually construct
this model and the key question I think
often is

if you have a particular phenomena or a
particular task in mind that you want
model to perform it's thinking well how
would I go about actually constructing
the data that's presented to a
participant performing that task
all presented to an agent and that's
trying to perform this task

and if you can explain how those data
are generated then you effectively have
the generative model that that
implicitly your agent or your
participant or whoever else should be
using to be able to perform that task
so really all you need to be able to do
is to work out how to create the

experimental stimuli and once you've
done that you've almost already created
your model
and the rest comes from
predefined pre-existing inversion
schemes that actually minimize the free
energy for that model
um so at that stage it becomes
relatively straightforward I say
relatively because as anybody who's has
experience encoding knows that a lot of
time is spent debugging and
throwing our barriers and spending a
while trying to work out why with a bit
of Googling to work out a solution
around that
um and I think those are all important
parts of the process
awesome Ollie or Terry on that recipe
and modeling question any thoughts on
that
um yeah uh actually
uh I also wanted to ask if
um uh do you have a plan for uh
extending the uh codes or supplementary
materials to include other languages
such as
imdp packages or
um any other packages that might come up
later and would be widely used in active
inference because as far as I know uh
there isn't any specific GitHub repo or
something dedicated for this book so far
so do you have any plans in developing
such a supplementary material for the
book
we don't at the moment
um it might be an interesting thing I

mean it might be a useful thing to for
to be done by by the actual Insurance
Community as well as a way of sort of
being engaged involved in it

I suppose one of the difficulties of
doing that for a book is that often
these things move on so quickly by the
time it comes out in a book that those
packages have moved on
um so I think you know it was it was
useful for us to give a simple example
of Matlab implementation to demonstrate
the principles in in one of the packages
that we've been involved in developing
as part of the SPM package where we've
been involved in developing the active
routines as part of that

um
and

I think it you know it's it's
certainly a good idea thinking about
having other

um uh repositories out there with all
sorts of demonstrations so the sbm
package does have a range of different
demonstrations for different sorts of
um uh implementations as well for
different kinds of models which is a
really helpful resource and I I often
suggest to people one of the first ways
of getting a sense of how to build these
models is to look at one that's already
been built and and do their best to
break it

change things and make it behave in
different ways

um and so I think you're absolutely
right to say examples of the code and

how it's implemented are vital in

bringing people into this

um but yes I think you're absolutely

right to say that it'd be really helpful

to have both you know Python and Julia

implementations and and all sorts of

different ways of doing it and examples

so that people can pick those up I agree

completely

and of course one of the advantages of

python and Julia here is is that they're

open source and you don't necessarily

need academic license to use it which

again I think increases the number of

people who can access it

giving an overview to someone who might

not have been following in the blow by

blow of our little literature Corner

over the last few years

how would you say active inference

modeling has evolved during your time in

the space where are we at just generally

speaking with applications what

applications are developed or possible

today that when you were starting your

PhD seemed like they were just

implausible

um

so I think

if we go back a bit I suppose some of

the earlier implementations were almost

all in the continuous domain

um so they're all based upon

um things like the generalized filtering

type algorithms that

um

were initially developed as sort of

filtering out

um methods to perform time series

inference

um those things then

equipped with an active component then

then the canes and the earlier active

inference type simulations and a lot of

that focused on things in the movement

domain whether that be eye movements or

um or control of Limbs to the extent

that you can sort of develop handwriting

and those sorts of things

one of the interesting things that

started to happen based upon that was to

think about how how you then get

something that looks more sequential

emerged from those continuous Dynamics

and handwriting was actually a really

good example of that where there were

some simulations developed in in which

um

in which the continuous model

um effectively had something a bit like

a logical Terror dynamic in it and the

idea bit there being that you can sort

of

um synological Terror Dynamics are a

kind of dynamical system often used for

modeling things like Predator prey

interactions and the idea is that you

get a peak in terms of a predator

population sorry prey population

followed by peak in the pre the

Predators as they eat the prey causing

drop in the prey population but then as

the Predators die out of the prey come

back you get this sort of sequence of

Peaks from Predator prey Predator prey

and you can generalize that to then have

you know multiple different populations
each praying on one another so you get
this sequence of Peaks over time
and that sort of emergence of sequences
from continuous Dynamics I think was one
of the key moves early on in terms of
thinking about things that are a bit
more cognitive in terms of how do I do
this and then this and this and start to
plan and generate much more purposeful
type of behavior
so that that went so far until
um until things moved much more to a an
explicitly discrete or sequential style
of modeling that's when things like
Markov decision processes it started to to
um be employed in the active entrance
domain or partially observed Markov
decision processes so those are models
that say I assume my world is a sequence
of States over time uh one after another
Each of which generates some
observations and from those observations
I can make inferences about the states
you can then say well if I can act upon
the world I can change the sequence of
States
um as they evolve over time
and by selecting different actions I can
or different ways I can change that
sequence and effectively planning and
making decisions so that's again where
you get into a much more cognitive sort
of domain
um some of the key advances
um sort of early on while I was doing my
PhD with the emergence of things like
deep hierarchical

models of things like Markov decision processes so here the idea was that in the world around us there are lots of things that evolve over different sorts of time scales

and so

um

one of the the early examples of this was was reading the idea that when you move your eyes around the page you're sort of looking at an individual word or the letters within that word that that word is itself part of the sentence and the time it takes you to infer what the word is is going to be much shorter than the time it takes you to work out what's going on in terms of the sentence the sentence and the paragraph and the paragraph on the page and the page in the book

Etc and so there's a whole hierarchy of different time scales in the world around us

and one way of accounting for that relatively simply in these sorts of models is to Simply stack them on top of one another so the outcomes you're now predicting from a from one of the mark of decision process type models I was describing

um now reflects a sequence of very fast States and each of those States yes exactly is the the graphic of this and then each of those

um outcomes predicts another sequence of very fast States so what you're left with is at the very top level a very slow progression of state after state

after state

for each of those states which might be
the sentence you then have the word the
words in that sentence beneath it and
then for each of those words you might
have each of the letters in that
sentence oh sorry in each of the letters
in that word

and

and so that then sort of expanded the
range of things that we can do in terms
of how we

um

give some sort of deep temporal
structure into the kinds of models we
develop

sometimes people refer to that as kind
of breaking the Markov property so for
those who who aren't familiar with this
the Markov property is the idea that
um that

each step in time depends only upon the
step immediately previous to it that
there's no sort of that all you need to
know about the next time step is what's
happening in the current time step
nothing else in the past is is useful
once you know that

um and people often suggest that means
that these processes are effectively
memoryless because it's only where I am
now that matters is not what was
happening well into the past by having a
hierarchical structure where you have a
higher level stage that is more slowly
changing that's contextualizing what's
going on at the lower level it means
that the fast

sequence of changes is actually
dependent upon something much slower
that's happening and that in itself is
dependent upon something much slower and
that gives you this a sort of memory it
allows for the Markov property to be
broken

foreign

so like I said we're sort of starting
off with continuous the emergence of
discrete from continuous the explicit
modeling of continuous models decision
making and the sort of exploration and
exploitation type trade-offs that you
get once you you treat planning as being
an influence process

the idea that you can then stack these
things hierarchically and have a range
of different time scales that break the
Markov process allow us to think about
things like memory and working memory
and I suppose one of the next stages was
the combination or the reintroduction of
the continuous models as the lowest
level of one of these hierarchies the
idea that

actually it's all very well being able
to make decisions but at some stage you
need to implement those decisions in a
world that's made up of physics and
muscle lengths and

um continuous sensory inputs and so
putting these sort of Hybrid models
together where you can use your
decisions to then predict short
sequences and trajectories defined in a
continuous domain

and I suppose since then they've you

know been a range of further
developments in action inference in that
setting
um and a range of developments
in in a number of other directions
um sort of in parallel
so some of those developments have been
thinking about implementation in other
languages some of those developments
have been
thinking about combination with methods
from Deep learning and instructional
variation a lot of encoders and function
approximators and amortized inference
um which a lot of people in the robotics
Community have been pioneering
particularly
um and
so that would be one One Direction that
things have been developed in and and I
suppose the other direction that's seen
a lot of development recently it has
been the sort of underlying principles
the um the physics that underwrites act
again for instance the principles and
how how you justify the sort of Dynamics
and where
where free energy comes from as a as a
concept and why it's useful in
understanding how um systems behave
so that I hope that wasn't too much all
at once but there was a sort of run
through of some of the developments that
um that spring to mind over the last few
years
awesome
Terry or Ali
any comments on that

quick historical review

oh again yeah I had another question
regarding

um actually it's a question

curiosity uh did you have any specific
reason for

um adopting this particular mathematical
notation throughout the book because
um

it seems like it somewhat diverges from
the notations used in some of the
well-known papers uh in the literature
and uh but I know uh active inference
notation uh is not

consistently used across the literature
but even in the papers that came out
after the publication of this book

written by Professor person or others

um I don't see this specific notation
used a lot in those papers so I just
wanted to ask about the reasons behind
this choice

foreign

I think you're absolutely right to say
that it's not always been used
consistently

um and it's worth saying that in some
there are a couple of different tensions
that that affect this so one is the idea
that in a lot of papers that are
published

um because it's only dealing with an
aspect of active influence and not
trying to to try to put it all together
they can sort of use a notation that may
be the same as notation you might use in
another field

um without worry that people will will

get conflicted and misinterpret one variable for another because everything's just defined within that one domain and another aspect to it is that in depending upon the paper in question and the audience is aimed at sometimes the notation is designed to um to match what what might be used in that broader field um whereas we've tried to try to make it as consistent as possible across different fields

um were there any particular notations that you were thinking help clarify uh yeah so for example um as just a very simple example um

using

ETA for external States instead of X that's been used or S and a for um active States and sensory States or other kinds of uses of lowercase matrices and uppercase matrices which is used differently in a paper such as a step-by-step tutorial on active inference and these kinds of notational divergences

okay um so yes I mean there's some quite nice illustrations so things like a and S which as you say are often used in some of the more physics orientated papers

um

unfortunately if you use those in a mock-off decision process paper people will interpret those as being States and

parameters of a likelihood Matrix

so unfortunately we couldn't use those
same letters without causing more
confusion

so what we try to do there is to to
focus more on the um sort of practical
inference side of things of you know the
actual implementations we tend to use of
active elements in particular problem
settings and there when you're
constructing a generative model
um particularly if it's in the
continuous domain we'll often use x as
being the states that I have to infer
which from the perspective of the
physics type papers are the external
States the things that are having an
effect on my sensory States

um we we've

I can't remember whether we used y or or
or o for the sensory States or the data
in the book and it's why as you think is
why okay

um and that that sort of fits with the
idea of using it as in the continuous
domain where in

um in a lot of settings when you're
trying to predict

um sensory data often why will end up
being the variable at the end of Newton

um so I I appreciate that it can be
confusing sort of jumping between
different resources

um but what we stroke for in this one
was internal consistency as much as we
could and trying not to overload
specific variables

or symbols

thank you

thanks

bigger 4.3 we bound ourselves coming
back to again and again it four sages
chapters seven and eight

and it lays out discrete time on top and
continuous time on the bottom in terms
of their structural similarities but
also they have some really important
differences so could you just as you see
it how

do the discrete time and the continuous
time models

have similarities and differences such
that you can lay them out in a
graphically or visually similar way yet
there's also some key differences for
example in what each of the nodes means
so how are they similar and different
and what did you hope to illustrate by
laying out figure 4.3 this way

so I suppose the reason that they can be
seen as similar is that effectively what
you're trying to do in both cases is to
represent a trajectory we're trying to
represent how something evolves in time
and that's something maybe the state's s
in the discrete time or they may be the
trajectory of some in States X in the
continuous domain

and when you're trying to say how do I
actually

put together a set of numbers that tell
me how something's evolved or evolving
there are a couple of different ways you
can do that one is to say I'm going to
tell you where it is at time one and
then I'm going to tell you where it is

at time two and then at time three etc

etc

and that's a perfectly legitimate and
perfectly reasonable way of writing down
a trajectory

it obviously misses out all the times
between time one and time two which may
or may not be significant depending upon
what the problem is you're dealing with
uh

the second way that we approached is to
say well you can use what what's
known as a Taylor series approximation
or a Taylor series or a polynomial
expression or that trajectory

um so essentially what we're saying
there is that you just construct
a function that is that that follows
that the the the shape of that
trajectory over time and you can
construct that function by saying well
let's look at where I am now at this
point in time and my position
how am I changing at the time if you
just took those two variables together
you just have a straight line telling
you where you are and where you're going
to be

if you then take account of your
acceleration you may even now have a
curved line which says if I'm here now
I'm going to be there then

um and as you add in more and more
elements to this so the rate of change
of the velocity and the rate of change
of the rate of change of the Velocity
Etc and just add in more and more rates
of change you effectively get a more and

more precise function telling you at least locally around where you are now where you're going to be at various other times

um so that that difference the either using Taylor series coefficients which are just

um successive rates of change or by using a sequence of States over time are what's Illustrated in in the lower and upper

um Graphics here respectively

um each of those things has there as you said that they have a lot of in common in terms of their sculpture in terms of the num or the number of numbers you need to be able to describe them

um but it has their advantages and disadvantages depending upon what you're trying to actually model

and I think for me I tend to think of the the sequence of categorical States over time as being quite useful in thinking about anything to do with decision making deciding between Alternatives

um thinking about sequences like language and words and or even sequences in terms of the eye movements you make looking in one place then the next and the next rather than continuously moving your eyes around space in a smooth fashion

uh whereas the process of interpreting sensory data that's coming in from your retina or from your your stretch receptors or whatever else or in terms of actually driving action that changes

continuous variables in the world there I think is much more useful to be able to describe things in terms of a continuous number and those sorts of continuous trajectories that you can get through using these Taylor series type approximations

which often referred to in this context as generalized coordinates of motion awesome

cool well another figure that we come back to again and again tucked away in chapter nine but giving a lot of context is uh figure 9.1

on what is termed meta Bayesian inference so could you describe what is happening in this figure

and is this exactly as every empirical ethologist is going to agree with is this controversial is this novel so what is it doing for octave inference when we conceptualize our

empirical task

structurally like this and do you think that that is exactly how ethologists have long seen this topic or is this something contentious

I think so I'll I'll talk through the figure but then but then you can tell me if that's how ethologists traditionally look at it and whether whether it's contentious I'll be interested to um

so the the idea behind this figure is to to highlight that when when dealing in psychology or Neuroscience or Fields where you're trying to understand um

our particular system is behaving or how

it's

um

essentially what's going on in the mind

or the brain or the nervous system of of

some creature that is behaving in some

way

um there are two levels of which you can

look at it one is saying well from my

perspective as the experimenter

what are the data I'm measuring and that

may be behavioral measurements from that

creature it may be taking

electrophysiological measurements it may

be

asking it a question if it's a patient

in a clinical setting

um and

I'm essentially now trying to optimize

my model of this system

now

in the neurosciences and in the settings

I've outlined above the the

um the kind of system we're interested

in is the sort of system that has its

own model

of other sorts of system

and so the kinds of questions we're

interested in the kind of parameters

we're interested in are things like the

prior belief that this creature has

about this thing

or how precise it thinks it's data are

and so the parameters I'm now asking a

question about are the parameters of the

beliefs of another of another system

so the outside box here

um the the one that the parameters are

going into the experimental stimuli are
going into and The observed Behavior
coming out of
um is essentially my model of how this
other system behaves
now the inside box there the one in the
dotted line or the dashed line
um that now represents the model that
that system has
to understand the world around it
um and so when when we construct our
model that contains another model inside
it
um and by saying if this creature
believed this this is how they would
make inferences about their environment
and generate Behavior I can now predict
the behavior I would observe if this
creature
believed this thing
um so the subjective objective idea is
that the objective model is is the
experimenter
um and its beliefs and the experimenters
beliefs and the observations they make
by measuring what the participants or
experimental subject does
and the subjective model is the
participants model of the world around
them
so that I think would be my summary of
how this figure works and the the idea
it's trying to get across now you can
tell me how ethologists
May disagree with that or agree with it
or what they'd find controversial
I'll give a thought and Ali or Terry
feel free to give a follow-up or a

thought

I don't think in principle it's

contentious or dissentious

however there might be an explicit or an

implicit belief

course ironically that we can simply

take

some of these parameters coming into our

experimental setup

and even the selection of experimental

stimuli we can take those as simply

given which is a very frequentist

perspective and then on the observed

Behavior we can simply use descriptive

statistics

we'll characterize the predefined groups

according to the one experiment we ran

we'll do the t-test we'll do the Anova

we'll come up with a p-value it's going

to be frequentism all the way forward so

even if we're doing frequentism about a

generative model

there's a sense that if we just clamp

down on our experiment and treat it as

handed down without error instead of

being actually drawn from distributions

of possible experiments we could have

done

if we clamp down Upstream of the

experiments we'll just be able to

describe and ultimately publish without

needing to look at what else could have

happened

and I think this diagram and also the

very subtle dashed line around even the

bigger box

suggests that just like the active

clinician engaging with a conversation

or the active experimenter also engaging
iteratively over their system of
Interest
that itself
is an Adaptive epistemic process
where
yes you could simplify it by treating
the parameters and the stimuli sequences
as simply given but then you don't have
access to the generative model that
actually is the generative process for
the subjective model
so summarize in principle uncontentious
however sometimes discordant with the
ways that behavioral experiments are
conceptualized planned and analyzed
one reason why it's helpful because
during the book we we try to stay on the
road on the high road of the low road
but
um
when we're in discussions
it's almost like there's a whole road
map and people have their own
backgrounds and questions and it kind of
comes together with all these different
um
topics that people are curious about
ultimately that bring them to active
inference so
going to this metabasian figure
helps in some ways situate
how the authors are seeing this entire
process
of
experimental design observation and
cognitive modeling and that was unpacked
also in the recent map territory fallacy

fallacy work

Ali or Terry

any

thoughts or questions in our last 10 minutes

is there a figure you want to look at or we can go to some other places maybe we can ask the missing figure in uh it was in I think it was in figure seven point

yes yes or something 7.2 in the lower left

so then this was actually corrected by I I don't know quite how it ended up Vanishing but it was corrected in one of the later print versions I'll see if it's in the version of course in front of me now and I can show you what it would have looked like

um but apologies for that uh which figure is it

7.2 on page 128. and

um unfortunately I picked up a copy of the book that doesn't have it

um so there have only been one and a half fractal dimensions of printing yes

um yeah I think unfortunately I don't think that one's corrected yet in the online copy

um essentially what would have appeared in that that figure or what did in some of the later printing versions is um

the errors that are driving the updates of the states in the plot above

um now because the

so the errors here are um are

effectively the

um are the free energy gradients so uh

or the negative three energy gradients

so essentially the

um

as you as you go down the free energy

gradient you're effectively minimizing

your error

so what that means in practices that the

errors would effectively be the rates of

change of the states and the plot above

so if you take the states and the plot

above and you imagine what the rates of

change of each of those lines are that's

what you would get in the plot below

well on this same kind of Topic in

chapter seven we're in the discrete time

generative models and um

a little bit less so here but in the

latter models

here

in 7.13

and in what we just looked at in 7.2

where are these continuous

interpolations coming from

if it's a discrete time model

that's a good point so I I suppose

there's an important distinction between

if there's there's the distinction

between

um the things that are being inferred

which are happening over discrete steps

in time and then there's the inference

process itself

um which deals in probabilities which

are themselves continuous they can vary

in this case anywhere between zero and

one

um and the time over which that
inference happens

so

to give you an example if you imagine
that you're you the way you're analyzing
this figure at the moment is by making a
series of category movements and you're
not you're not sort of continuously
moving your eyes from one block to the
next you're making a series of rapid
um ballistic movements that
um and take you from one to the next
um and effectively every time your eyes
stand still you have a new data point
that you're now trying to analyze
you can imagine that that between that
or every time your eyes stays still
everything else in your brain doesn't
just stop

um so once you've got that new data
point your brain now has some time to
simulate that data point to draw
inferences about what is seeing there
and those inferences happen through the
continuous dynamics of what's happening
in in your neurons and various networks
across your brain

um so you can think of the continuous
changes that are happening for each
state as as not being changes in the
world that we're trying to infer not
being changes in the data point I'm
currently looking at the changes in my
beliefs about it before I then sample
the next data point

um

interesting

I have a few more figure based quick

questions but Terry do you want to bring up anything or just give any remarks no worries if not well I I am very much just trying to understand these Concepts so I I don't make any pretense uh having any deep understanding but the the key thing that interests me is how you jump from this sense of uh how we update our predictions into the clinical realm and I I I'm interested in pain medicine and I see neuroplastic pain is very much a um mistaken prediction based on often uh misinterpretation by medical professionals and the information that they give to patients and I'm really intrigued by the potential for uh using these Concepts to to give patients an architecture in which they can start to understand that they can that they have some agency and altering their future predictions and really change in their experience of their pain and and as you're a clinician I'd be really intrigued to get your sense of where this is going in that type of print clinical Realm um I think that's a really nice example as well thinking about um thinking about attitudes was pain and particular kinds of pain particularly when there isn't a sort of obvious tissue damage that can

be identified that's um that's causing
that pain that still is obviously a very
real sensation to the people who are are
experiencing it

um and I think it generalizes to a
number of other other conditions
um and probably there's an overlap with
things like functional neurological
disorders probably a number of medically
unexplained symptoms and things where
that there is a huge amount of overlap
in these sorts of processes where you
can't necessarily directly identify a
big structural lesion of any sort but
clearly the way the nervous system is
working

um something isn't quite isn't quite
right

um pain I think is an interesting
example particularly because
pain is never a real thing out there in
the world is it I mean it's it's always
an inference and so I think that's one
of the first things to say to anybody
you know regardless of whether he got
pain because you've lost a leg or
um whatever else it there's no such
thing as pain as an external state it is
just your inference your explanation for
the the

um signals that are coming in from your
body

um

now one of the important Concepts that I
think comes into all of these

um sorts of syndromes and probably many
others as well is the idea of precision
the idea that

um the data I predict sensory data
coming in

um I I can in addition to estimating
what those data are going to look like I
also have to estimate how confident I am
in what they're going to look like
um if I'm not very confident or I think
I'm dealing in something that's very
noisy then I can actually discount a lot
of that

whereas if I believe that it's all
extremely reliable very clean data
that's coming in then I want to put a
lot of weight on that
and another way of saying that is that
there are some things I pay attention to
and some things I don't

um

and

you know just saying it's reliable data
is just a matter of saying I'm going to
attempt to it

now it might be that in a number of
these different syndromes

um what's going on is that our bodies
have

um overestimated the Precision
overestimated how clean the data are
coming in from a particular body part
um and that that's why we then over
interpret any signals that are coming in
as if they are painful and the
influences we draw from those signals
are just too confident and too driven by
something that they shouldn't be
an example I often like to use in in
clinical discussions is that it I told
you to imagine to pay attention now to

the the feeling that your left sock is
having on your left big toe you can
probably feel it
um but you were completely unaware of it
before but now if I say continue to pay
attention to that and carry on paying
attention to that and and you're unable
to pay attention away from it anymore if
you felt that all the time it would feel
unusual it would feel wrong
um and
um maybe a misestimation of precision in
that way maybe what's going on in a lot
of these sorts of conditions
partly I think it's often interesting
that in in some of these people there's
um there was at some stage an injury or
something that affected that body part
and you can imagine that if you went
through a period of illness where when
you did have tissue damage and we're in
pain for that reason it might direct
your attention to that and you may have
a learned deployment of attention
towards that area
that might persist even after
um even after the the insults and the
damage was resolved
um and again maybe maybe it's that sort
of persistent uh mislearning of
um how precise things are that leads to
the continued experience of pain even
when there's nothing
um sort of externally causing it anymore
so I think just having a sense of
inference and of um
and the idea that that Allah
perceptions of the world around us are

ultimately inferences

um says that you know if we change our model in some way we're able to adapt and change and develop plasticity in our our brains and spinal cords can be exploited uh then then we can also um correct those mislearned models um and hopefully resolve those sorts of symptoms and syndromes

awesome

thank you well we are at the very close really appreciative Thomas of how much we were able to cover

what are your closing words or thoughts potentially to someone who has listened to this one hour discussion as their appetizer before joining the par at all restaurant service

um

well I know I've I've enjoyed the discussion thank you for having me here um

I that it sounds I mean I'll reiterate what I said earlier that I think one of the key things in in making this field more accessible to more people are these sort of

um uh reading groups the ideas of trying to create these educational resources and to unpack and discuss these things um and you know I'm impressed by all the infrastructure you've put together and um and by everything you've managed to achieve by doing this

um so I'd certainly encourage anyone who's interested to join up to your reading groups

um and I hope they they get something

out of it and out of the book

thanks Ali

well uh yes um actually since the

beginning of the first cohort

um this book has practically become a

part of our life that we love and

cherish and uh it's amazing to see how

with each iteration of our reading group

uh it gives rise to whole different set

of questions and and discussions and uh

actually I'm a firm believer in this

famous quote I don't know by whom which

says that the sign of a great book is

not how many answers it provides but

rather how many thought-provoking

questions it engenders so based on that

Criterion and judging by the variety of

questions uh brought up during our live

discussions as a kind of empirical

evidence for that I think it's probably

safe to say that this is truly a great

book so congratulations on that and

thank you

thank you very much

great closing thoughts so Thomas thank

you again you're welcome back anytime

thanks Ali and Terry for all the great

participation and until next time

bye